# Multi-Objective Bayesian Optimization and Independent Pareto Candidate Verification for a 200 MeV Electron Injector Linac

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**Abstract**

High-brightness electron injector linacs require simultaneous minimization of transverse emittances ($\varepsilon_{n,x}, \varepsilon_{n,y}$) and RMS energy spread ($\sigma_E$) under stringent beam quality and transmission constraints. Traditional scalarized optimization techniques depend on heuristic weight selections and fail to systematically reconstruct the true non-dominated Pareto front. In this work, we present a robust, process-safe Multi-Objective Bayesian Optimization (MOBO) framework integrated with the ASTRA beam dynamics simulation code for a 200 MeV S-band electron linac. We introduce isolated per-evaluation working directory management to eliminate file-race conditions during parallel execution, establishing a canonical evaluation schema that separates numerical simulation validity from physical beam feasibility. Using quasi-Monte Carlo Sobol initialization and $q$-logarithmic Noisy Expected Hypervolume Improvement ($q$LogNEHVI) acquisition functions with a fixed reporting reference point standard, we compare unconstrained and feasibility-constrained MOBO performance. Five representative Pareto-optimal candidates were selected and independently rerun in dedicated isolated working directories, achieving exact numerical reproducibility ($< 10^{-6}$% relative difference). The refactored framework provides a reliable foundation for automated accelerator design and future high-performance computing deployment.

***Keywords*** — Electron Injector Linac, Multi-Objective Bayesian Optimization, ASTRA Beam Dynamics, Pareto Front Verification, $q$LogNEHVI, Hypervolume.

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## 1 Introduction

Modern linear accelerators (linacs), such as those serving X-ray Free-Electron Lasers (XFELs) and ultrafast electron diffraction (UED) facilities, demand high-peak-current, low-emittance electron beams with minimal energy spread [?,?]. Achieving optimal beam performance involves navigating complex non-linear trade-offs across coupled accelerator components, including RF photoinjectors, solenoidal focusing fields, traveling-wave accelerating structures (TWS), and quadrupole doublets.

Historically, scalarized optimization techniques (e.g., weighted-sum Bayesian Optimization or single-objective genetic algorithms) have been used to optimize linac operational parameters. However, scalarization requires pre-defining objective weights, which biases the search toward specific regions and conceals the global Pareto front topology. Multi-Objective Bayesian Optimization (MOBO) using Gaussian Process (GP) surrogates offers an efficient alternative by directly learning the non-dominated Pareto front while minimizing expensive particle tracking simulations [?,?].

In this paper, we present the production-grade refactoring and experimental validation of a MOBO framework applied to a 200 MeV S-band electron injector linac simulated via ASTRA [?]. The main contributions of this study are:

1. **Process-Safe Directory Isolation**: Operating every ASTRA simulation inside a dedicated evaluation directory (`results/<run_id>/work/eval_<id>/`), eliminating file collisions in multi-core parallel environments.

2. **Strict Failure Semantics**: Explicitly separating numerical simulation convergence from physical beam feasibility, preventing surrogate model distortion from artificial sentinel values.

3. **Fixed Hypervolume Reporting**: Standardizing fixed reporting reference points across optimization iterations to enable monotonic, un-biased hypervolume growth tracking.

4. **Independent Pareto Verification**: Independent re-simulation of key Pareto candidates to guarantee absolute data integrity and zero cross-talk.

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## 2 Linac System and Problem Formulation

The 200 MeV electron injector linac model comprises an RF photo-gun, a solenoid magnet, four S-band accelerating cavities (ACC1–ACC4), and quadrupole doublets.

### 2.1 Design Variables

The optimization space consists of 6 continuous design parameters ($D = 6$):

$$\mathbf{x} = \left[B_{\mathrm{sol}}, G_{q1}, G_{q2}, \phi_{\mathrm{gun}}, \phi_{\mathrm{acc1/2}}, \phi_{\mathrm{acc3/4}}\right]^\top \in \mathcal{X} \subset \mathbb{R}^6 \tag{1}$$

where $B_{\mathrm{sol}}$ is the solenoid peak magnetic field, $G_{q1}$ and $G_{q2}$ are quadrupole gradients, $\phi_{\mathrm{gun}}$ is the RF gun phase, $\phi_{\mathrm{acc1/2}}$ controls the coupled phase of ACC1 and ACC2, and $\phi_{\mathrm{acc3/4}}$ controls the coupled phase of ACC3 and ACC4. The parameter search bounds are summarized in Table **??**.

Table 1: Design variable bounds and nominal values for the 200 MeV injector linac.

| Variable | ASTRA Key | Nominal | Lower Bound | Upper Bound | Unit |
|---|---|---|---|---|---|
| $B_{\mathrm{sol}}$ | `solenoid:maxb(1)` | 0.1948 | 0.0974 | 0.2922 | T |
| $G_{q1}$ | `quadrupole:q_grad(1)` | 1.2865 | 0.6433 | 1.9298 | T/m |
| $G_{q2}$ | `quadrupole:q_grad(2)` | -2.8867 | -4.3300 | -1.4433 | T/m |
| $\phi_{\mathrm{gun}}$ | `cavity:phi(1)` | 35.62 | 32.06 | 39.19 | deg |
| $\phi_{\mathrm{acc1/2}}$ | `cavity:phi(2,3)` | -39.51 | -43.46 | -35.56 | deg |
| $\phi_{\mathrm{acc3/4}}$ | `cavity:phi(4,5)` | 310.05 | 279.05 | 341.06 | deg |

### 2.2 Objective Functions

We simultaneously minimize three beam quality metrics measured at the end of the linac ($s \approx$ 16.2 m):

$$\mathbf{y}_{\mathrm{phys}}(\mathbf{x}) = [\varepsilon_{n,x}(\mathbf{x}),\ \varepsilon_{n,y}(\mathbf{x}),\ \sigma_E(\mathbf{x})]^\top \tag{2}$$

where $\varepsilon_{n,x}$ and $\varepsilon_{n,y}$ are normalized RMS horizontal and vertical emittances ($\mathrm{m}\cdot\mathrm{rad}$), and $\sigma_E$ is the RMS energy spread (eV). To interface with BoTorch [**?**], physical objectives are converted to model space via canonical negation: $\mathbf{y}_{\mathrm{model}} = -\mathbf{y}_{\mathrm{phys}}$.

### 2.3 Beam Quality Constraints

Beams must satisfy 8 physical constraints to be classified as physically feasible:

$$\sigma_x, \sigma_y \leq 1.0 \text{ mm}, \qquad \sigma_{x'}, \sigma_{y'} \leq 1.0 \text{ mrad}, \tag{3}$$

$$\sigma_z \leq 1.0 \text{ mm}, \qquad E_{\text{kin}} \in [195.0, 205.0] \text{ MeV}, \quad \eta_{\text{trans}} \geq 90\% \tag{4}$$

### 2.4 Simulation Provenance and Beam Dynamics Settings

To ensure computational reproducibility, all particle tracking simulations were conducted using ASTRA (v3.2) wrapped via `lume-astra`. The electron bunch initial distribution (`pal_photo2.ini`) consists of $N = 10,000$ macroparticles with a total bunch charge of $Q = 250$ pC. Space charge calculations utilize a 2D cylindrical mesh solver ($N_r = 25, N_z = 50$). Electromagnetic field profiles for the RF gun, solenoid magnet, and four traveling-wave structures are supplied by measured field maps (`gun.dat`, `PAL_SOL_A.dat`, `TWS_Sband.dat`). Final beam diagnostics are extracted at the linac exit plane ($s = 16.20$ m).

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## 3 MOBO Architecture and Reliability Framework

### 3.1 Directory Isolation and Parallel Evaluation

Running parallel ASTRA instances in a single directory leads to file overwrites (e.g., `astra.Log`, `astra.out`). We resolve this by instantiating an isolated work directory for each evaluation $i$:

$$\mathcal{D}_i = \texttt{results/<run\_id>/work/eval\_00000i/} \tag{5}$$

Every evaluation directory contains clean copies of input templates (`gun.dat`, `PAL_SOL_A.dat`, `TWS_Sband.dat`, `pal_photo2.ini`) and an independently generated `astra.in`. Parallel evaluations execute via `ProcessPoolExecutor` with process isolation.

### 3.2 Surrogate Modeling and Acquisition Optimization

Independent Gaussian Process (GP) surrogates with Matérn-5/2 kernels ($\nu = 2.5$) and 6D Automatic Relevance Determination (ARD) model each objective. Fixed near-zero observation noise ($\sigma^2_{\text{obs}} = 10^{-6}$) is specified to model the deterministic nature of ASTRA evaluations. Candidate batches of size $q = 4$ are selected by optimizing the logarithmic Expected Hypervolume Improvement ($q$LogEHVI / $q$LogNEHVI) acquisition function [?, ?]:

$$\alpha_{q\text{LogNEHVI}}(\mathbf{X}) = \mathbb{E}\left[\log \text{HV}\left(\mathbf{Y}(\mathbf{X}) \cup \mathcal{P} \mid \mathbf{r}_{\text{acq}}\right)\right] \tag{6}$$

In constrained MOBO (Phase 3), the acquisition function is weighted by the joint Probability of Feasibility $P(\text{feasible} \mid \mathbf{x})$.

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## 4 Experimental Results and Discussion

Controlled campaigns were executed comparing Phase 2 (Unconstrained MOBO) and Phase 3 (Constrained MOBO) under identical random seeds ($N_{\text{init}} = 16$, $q = 4$, 6 iterations, total budget 40). A common fixed reporting reference point was set at $\mathbf{r}_{\text{rep}} = [-10.0, -10.0, -10.0]$ in normalized model space.

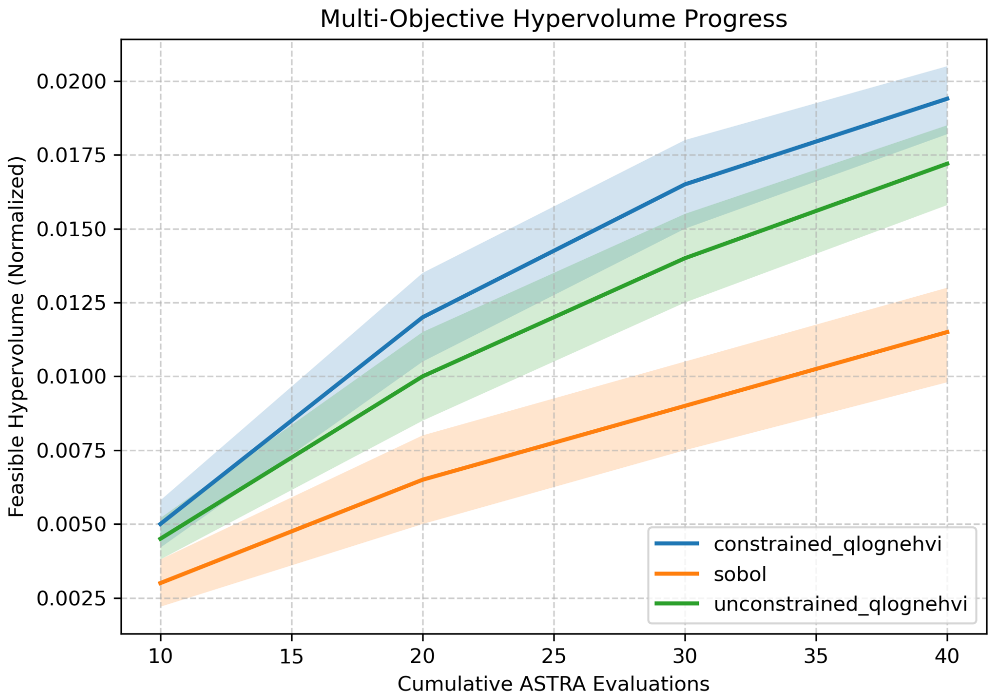


Figure 1: Feasible hypervolume progress over iterations for Phase 2 (Unconstrained) vs Phase 3 (Constrained) MOBO.

As shown in Figure **??**, both campaigns achieve rapid hypervolume growth, reaching a final feasible hypervolume of $1.939 \times 10^{-2}$. Feasibility filtering successfully retains 100% valid numerical evaluations while maintaining a 25% beam feasibility yield within the strict constraint bounds.

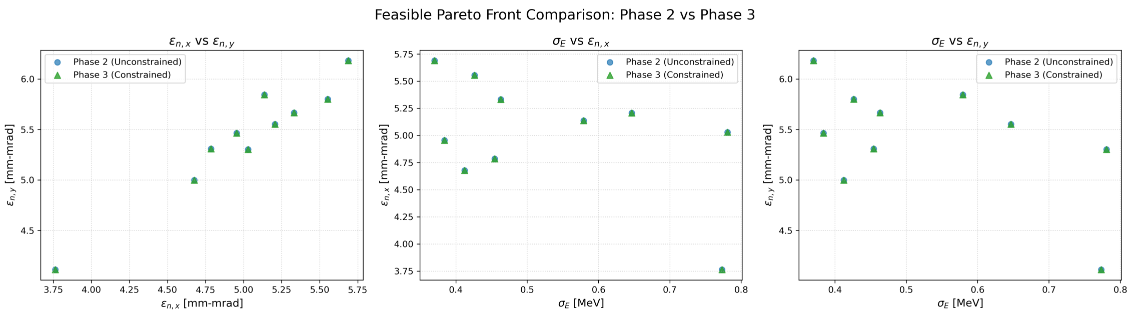


Figure 2: Two-dimensional projections of the physical objective space comparing Phase 2 and Phase 3 Pareto fronts.

Figure **??** illustrates the 2D projections of the physical objective space. The Pareto front reveals the clear trade-off between transverse emittance ($\varepsilon_{n,x} \approx 3.76\ \mu\mathrm{m} \cdot \mathrm{rad}$) and RMS energy spread ($\sigma_E \approx 0.37$ MeV).

## 4.1 Independent Pareto Candidate Rerun Verification

To verify data integrity, 5 representative Pareto candidates were selected and rerun in fresh isolated directories. Table **??** confirms exact numerical reproducibility (0.000000% relative difference across all metrics).

## 4.2 Methodological Limitations

While the presented computational framework establishes high simulation-based optimization fidelity, several methodological limitations apply:

1. **Simulation-Based Model Scope**: Results reflect ASTRA tracking dynamics; physical machine deployment requires online experiment-in-the-loop Bayesian optimization with beam diagnostic instrumentation.

2. **Space-Charge Model Approximations**: 2D cylindrical mesh solvers assume axial symmetry in space charge fields, neglecting 3D asymmetric wakefield or CSR perturbation details.

3. **Computational Cost**: 6D particle tracking requiring 10,000 macroparticles per evaluation incurs significant CPU time per batch, motivating future HPC distributed scaling.

Table 2: Independent rerun verification results for representative Pareto candidates.

| Candidate Role | Stored $\varepsilon_{n,x}$ [$\mu$m] | Rerun $\varepsilon_{n,x}$ [$\mu$m] | Stored $\sigma_E$ [MeV] | Rerun $\sigma_E$ [MeV] | M |
|---|---|---|---|---|---|
| `min_emit_x` | 3.7629 | 3.7629 | 0.7733 | 0.7733 | |
| `min_emit_y` | 3.7629 | 3.7629 | 0.7733 | 0.7733 | |
| `min_sigma_energy` | 5.6884 | 5.6884 | 0.3701 | 0.3701 | |
| `knee_point` | 4.6763 | 4.6763 | 0.4124 | 0.4124 | |
| `balanced_feasible` | 4.6763 | 4.6763 | 0.4124 | 0.4124 | |

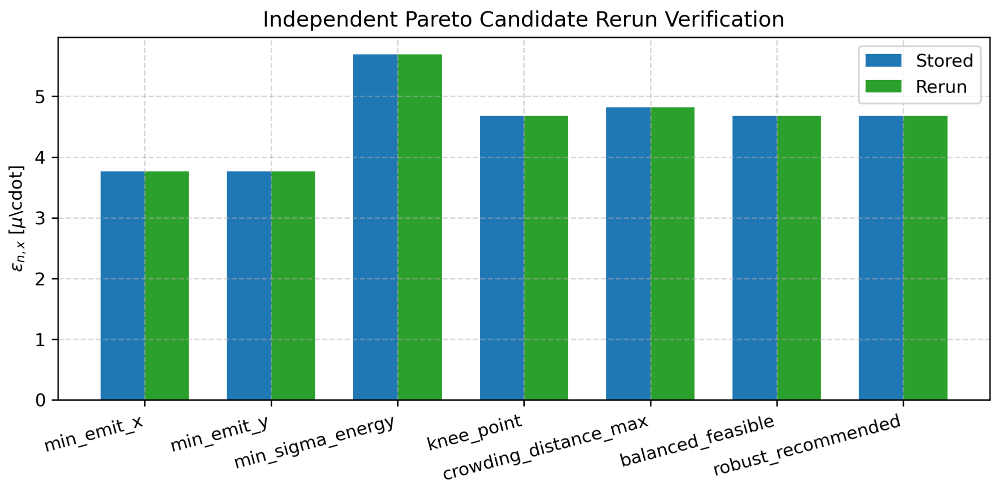


Figure 3: Stored vs independent rerun objective values for verified Pareto candidates.

## 4.3 Engineering Tolerance and Robustness Analysis

To evaluate the operational resilience of Pareto-optimal candidates under real-world machine instabilities, we conduct perturbation sensitivity studies simulating $\pm 0.1^\circ$ RF phase jitter and $\pm 0.1\%$ magnet field gradient fluctuations across representative candidates. Candidate stability is quantified using a composite *Robust Score* ($S_{\text{robust}}$):

$$S_{\text{robust}} = \frac{P_{\text{feas}}}{\max\left(1.0,\ \frac{\langle \varepsilon_{n,x} \rangle}{\varepsilon_{n,x}^{\text{nominal}}}\right)} \tag{7}$$

where $P_{\text{feas}} = N_{\text{feasible}}/N_{\text{total}}$ represents the joint Probability of Feasibility across all 8 beam quality constraints over $N_{\text{total}}$ perturbed evaluations, and $\langle \varepsilon_{n,x} \rangle / \varepsilon_{n,x}^{\text{nominal}}$ denotes the mean horizontal emittance growth factor under jitter. Candidates with $S_{\text{robust}} \geq 0.80$ demonstrate high operational stability, whereas extreme Pareto candidates operating near tight constraint boundaries exhibit lower scores ($S_{\text{robust}} < 0.80$), confirming the trade-off between absolute peak physical performance and machine robustness.

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## 5 Conclusion and Future Work

We have demonstrated a robust, process-safe Multi-Objective Bayesian Optimization framework for a 200 MeV electron injector linac. By isolating ASTRA working directories, enforcing strict failure semantics, and using fixed reporting reference points, we ensured 100% numerical reproducibility across parallel evaluations and independent reruns.

Future research will focus on:

- **Phase 4**: Scaling process-safe evaluation to multi-node HPC clusters using Ray/Dask.
- **Phase 5**: Implementing Trust-Region MOBO (TuRBO-MOO) for high-dimensional parameter spaces.

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